# Predicting Credit Card Fraud

CSC 3220 Final Project, Deliverable 1: Problem Statement and Data
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## 1. Problem Statement and Background

**Problem.** Every card transaction has to be approved or declined in a fraction of a second, and in that instant the issuer must decide whether the person holding the card is really the cardholder. Our project asks how well we can estimate the probability that a transaction is fraudulent, given only what is known at the moment it is made: the amount, timing, merchant, device, authentication method, and account history. The output is a fraud probability for each transaction, which a bank could then turn into one of three decisions: approve, decline, or send for review.

**Who cares and why it matters.** Fraud is expensive, and it keeps getting more so. The Federal Trade Commission reports that U.S. consumers lost a record $15.9 billion to fraud in 2025, and that cost does not land on one party. Cardholders lose money, time, and trust in their bank when fraud slips through. Banks and card networks usually reimburse those losses and then pay to investigate the resulting disputes. Merchants, for their part, are stuck with chargebacks and fees on sales that were never legitimate to begin with.

The opposite error is just as real, if less obvious. A legitimate purchase that gets declined (a false positive) frustrates the customer and costs the merchant a sale. A better model could catch more fraud without blocking more honest customers, or hand investigators a shorter and more accurate list of suspicious transactions.

This problem is harder than ordinary classification for two reasons. First, fraud is rare; in most real data sets, well under 2% of transactions are fraudulent. A model that labels everything "legitimate" would therefore score over 98% accuracy while catching no fraud at all, which is technically impressive and completely useless. Second, fraud patterns do not sit still, because criminals adapt. A recent example is the use of generative AI to write convincing scam messages.

**Related work.** Credit card fraud detection is a well-studied machine learning problem, and the literature points in a fairly consistent direction. Bhattacharyya et al. (2011) compared logistic regression, support vector machines, and random forests on real card transactions; random forests performed best overall, but logistic regression held up as a strong and interpretable baseline. Dal Pozzolo et al. (2015) released the widely used European credit card data set (0.17% fraud) and showed that class imbalance distorts predicted probabilities, so rebalancing the training data requires correcting those probabilities afterward. Chawla et al. (2002) introduced SMOTE, which creates synthetic examples of the rare class and remains one of the most common ways to handle imbalance in fraud work. More recently, gradient-boosted tree methods such as XGBoost (Chen & Guestrin, 2016) have become the standard for tabular competitions, fraud prediction included.

**Measuring success.** Because fraud is rare, accuracy tells us very little. We will instead judge success by how well a model ranks fraudulent transactions above legitimate ones on held-out data it did not see during training, using threshold-free measures such as the area under the ROC curve. We will also look at how much fraud the model catches at a reasonable false-alarm rate. Our measurable goal is for the primary model to clearly beat two baselines: predicting the overall fraud rate for every transaction (an ROC AUC of 0.5) and a simple logistic regression. Exact metrics and values will be reported in the Results section.

## 2. Data and Exploratory Analysis

**Dataset.** The data comes from the Kaggle competition Predicting Credit Card Fraud (Playground Series, Season 6, Episode 11), which is linked in the Appendix. Like all Playground Series data, it is synthetic; Kaggle generated it to resemble a real data set, Credit Card Fraud Detection 2026, so its feature distributions are close to the original's but not identical. The competition provides a training file of 699,960 transactions with the fraud label and a test file of 299,984 transactions without it. We are not submitting to the competition, and the test file has no labels, so we use only the training file. Part of it will be set aside as the held-out data described in Section 1, with the fraud rate kept the same in each part. Each row is one transaction, with an id and 24 predictor variables (Table 1). The target variable is is_fraud (1 = fraud, 0 = legitimate). Only 1.67% of the transactions (11,689 rows) are fraud, so the two classes are severely imbalanced.

Table 1: The 24 predictor variables, grouped by type. In variable names, txn means transaction. A number in parentheses is the number of categories.

| Group | Variables |
|---|---|
| Transaction | amount_usd, distance_from_home_km |
| Timing and velocity | hours since last txn, txn count last 24h, velocity score, time of day hour, day_of_week |
| Customer and account | customer_age, card_age_months, account_balance_usd, prior_disputes, cvv_retry_count |
| Merchant | merchant_risk_score, merchant_category (12) |
| Context (categorical) | card type (5), auth method (5), channel (5), device type (8) |
| Risk flags (true/false) | is_foreign_transaction, is_new_merchant, used_vpn, ip_country_mismatch, billing_shipping_mismatch, is_ai_generated_scam_attempt |

**Cleaning.** The data needed very little cleaning, and we kept every row.

*Missing values and duplicates.* The training file has no missing or blank values and no duplicated transactions, so nothing had to be removed or imputed. This is typical of synthetic data; real bank data would almost certainly need imputation.

*Data types.* The six risk flags were loaded as the text True/False and the five categorical variables as plain text, so we converted the flags to 0/1 numbers and the categorical variables to factors. That way, every model reads them the same way. The target was stored as a decimal number, so we also added a labeled copy (Legitimate or Fraud) for plots. Day of week (0 to 6) and hour of day (0 to 23) are stored as numbers but are really cyclical categories, and we will decide in Deliverable 2 whether to encode them as factors or as sine/cosine pairs.

*Outliers.* The two money variables are strongly right-skewed. The transaction amount has a median of $27.25 but a maximum of $6,873, and the account balance reaches $99,701. The 1.5 × IQR rule flags 70,741 amounts (about 10%) and 29,068 balances (about 4%) as outliers. We did not remove them, since they look like plausible large purchases rather than data errors, and large purchases are more often fraud (Figure 2). Instead, we added log-transformed copies of both variables, which reduce the skew for plots and linear models. Tree-based models are not affected by skew.

*Anomalies.* Two kinds of rows do not make real-world sense. First, 29,385 transactions have the ATM channel but a device other than an ATM machine (an iPhone or a smart watch, for example). Second, 7,470 transactions are for more than the account balance. These are probably artifacts of how the synthetic data

was generated. We kept both and added a 0/1 column marking each kind, so the models can learn whether they matter.

**R packages and code.** We loaded the data with base R's read.csv(), cleaned and summarized it with the tidyverse's dplyr, and measured skewness with the moments package; Figures 1 and 2 were drawn with ggplot2 and Figure 3 with base R graphics. The code lives in five scripts in our repository: three exploration scripts (00a_first_look.R, 00b_fraud_patterns.R, and 00c_consistency_checks.R), followed by 01_load_clean.R and 02_eda_figures.R. The comments in those scripts explain each step and what each package is used for.

**Exploratory plots.** Figure 1 compares the fraud rate when each true/false risk flag is present and when it is absent. Every flag raises the fraud rate, some by a large amount. A transaction with an IP-country mismatch is fraud about 16% of the time, roughly 10 times the overall rate, and foreign transactions, billing/shipping mismatches, and AI-generated scam attempts are each fraud 12 to 15% of the time. The number of failed CVV attempts (not shown) has the strongest effect of all: two failed attempts raise the fraud rate to about 42%. Unfortunately, most of these flags are rare. All except the new-merchant flag appear in fewer than 4% of transactions, so they can explain only part of the fraud.

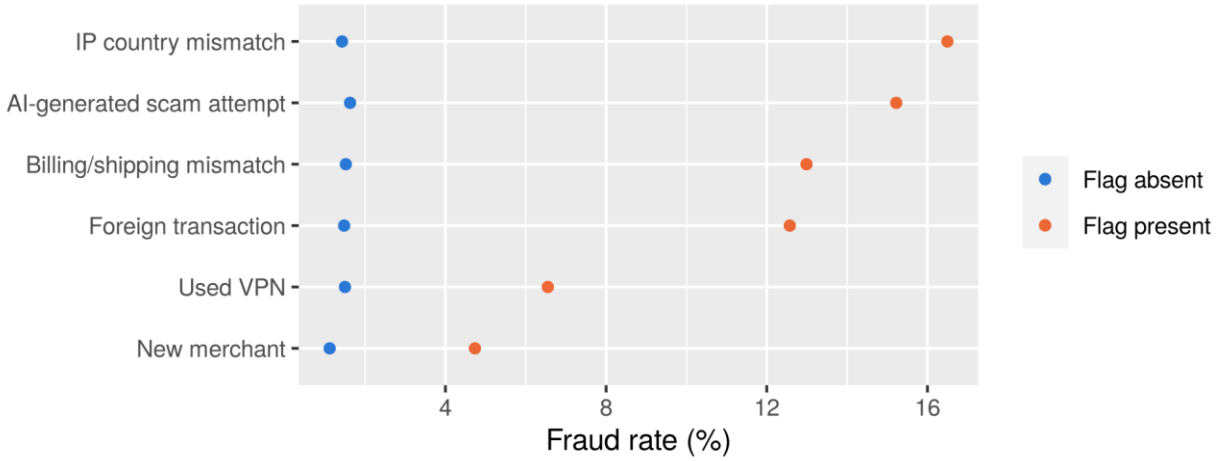



Figure 1: Fraud rate when each true/false risk flag is absent (blue) and when it is present (orange). The overall fraud rate is 1.67%.

Figure 2 shows the fraud rate for two categorical variables and for ranges of the transaction amount. Authentication method separates the classes clearly; transactions with no authentication are fraud 6.4% of the time, while biometric transactions are fraud only 0.5% of the time. Merchant category matters as well. Gift cards (7.3%), crypto exchanges (6.5%), and gaming (5.5%) have far more fraud than groceries or online retail (about 1.1%), which makes sense, since gift cards and crypto are easy to convert into cash. Fraud is also more common for purchases over $100. In contrast, card type, channel, device type, day of week, and hour of day (not shown) each stay within a few tenths of a percentage point of the overall rate. They may still help models through interactions with other features, but on their own they show little signal.

Figure 3 compares the distributions of six continuous variables for legitimate and fraudulent transactions. Fraudulent transactions have clearly higher merchant risk scores (median 45.9 vs. 34) and velocity scores (26.6 vs. 18.5), which fits the idea that fraudsters target risky merchants and make many purchases quickly once they have a card. Fraud victims also tend to be older (median age 61 vs. 52). This could reflect fraudsters deliberately targeting older customers, and we intend to look into it further. Distance from home and card age, on the other hand, look almost the same for both classes, and so does account balance (not shown); their correlations with the fraud label are all below 0.01. Even the strongest continuous predictor,

merchant risk score, has a correlation of only about 0.13. No single variable separates the classes, so we expect models that combine many weak signals, such as tree ensembles, to do better than any simple rule.

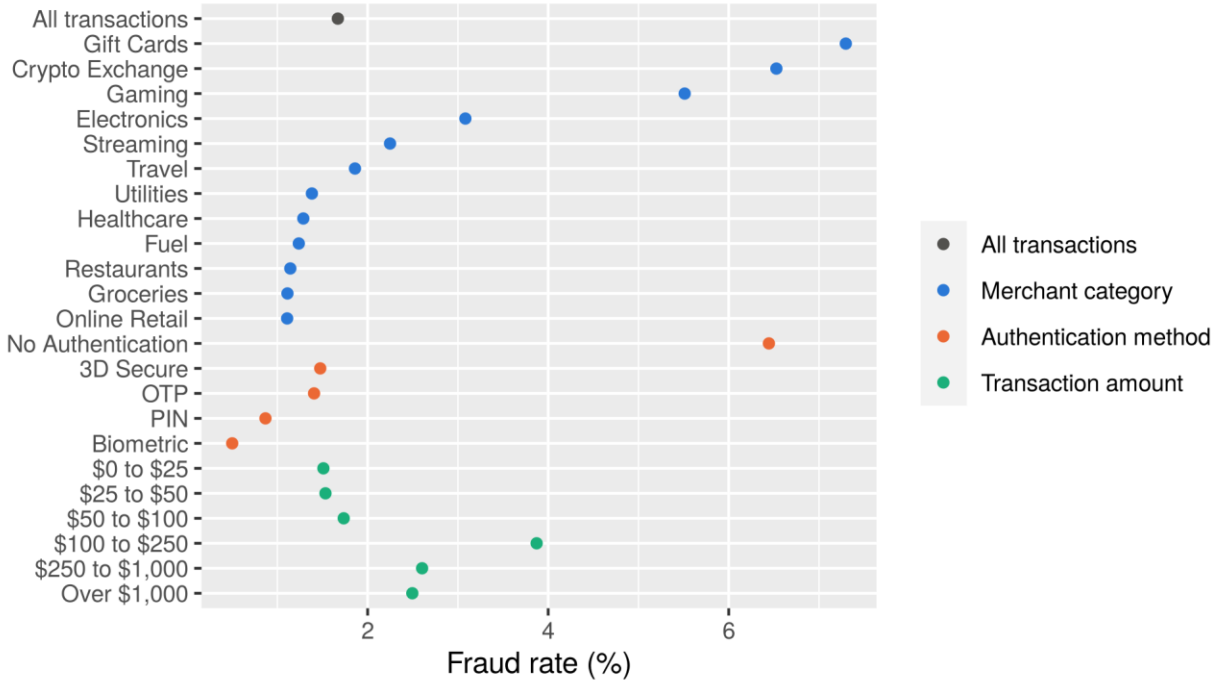



Figure 2: Fraud rate by merchant category, authentication method, and transaction amount. The gray dot at the top is the overall fraud rate (1.67%).

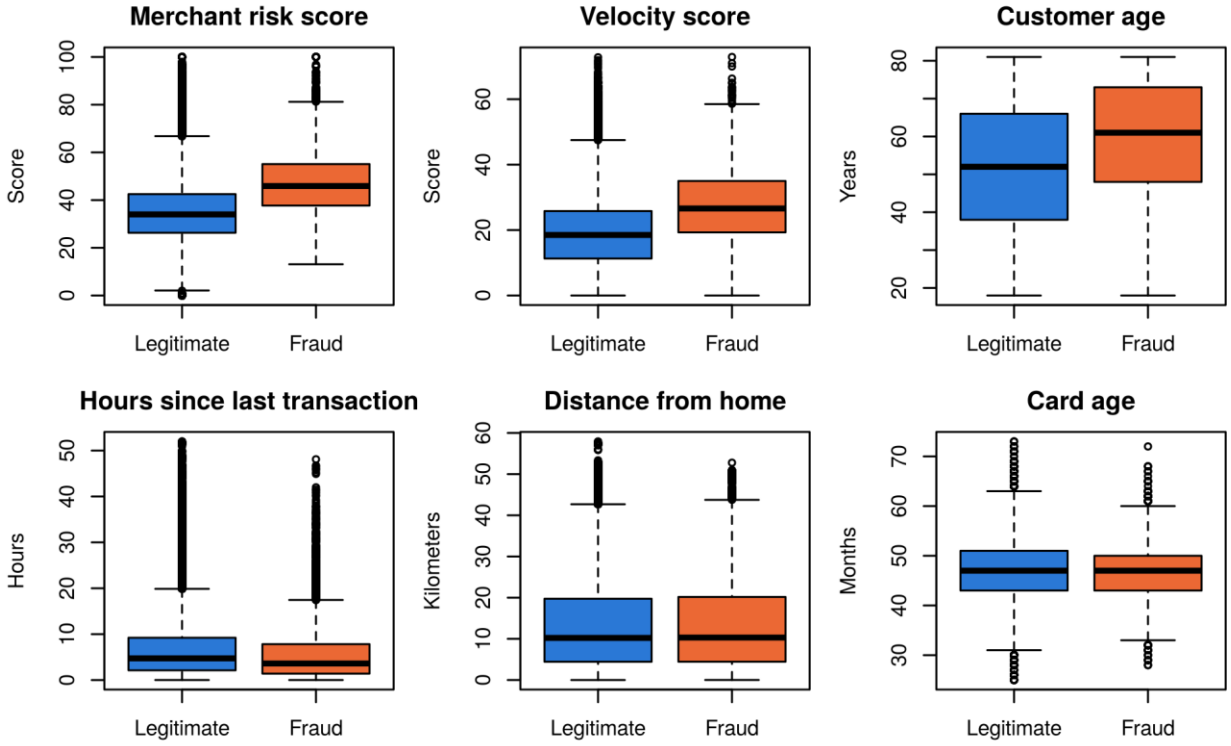



Figure 3: Boxplots of six continuous variables for legitimate and fraudulent transactions. The thick line is the median, the box holds the middle 50% of values, and circles are outliers beyond 1.5 × IQR.

**What we learned.** The data is clean, with no missing values or duplicates, but it is heavily imbalanced, so we must evaluate with ranking-based metrics rather than accuracy. The strongest signals are the rare risk flags (failed CVV attempts, IP mismatch, foreign transactions), authentication method, high-risk merchant categories, merchant risk score, and velocity score. Many other variables, such as card type, device, channel, and timing, show little signal on their own. Finally, the money variables are heavily skewed and some rows defy real-world logic; we kept those rows, transformed or flagged them, and will check whether the flags help the models.

**References**

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